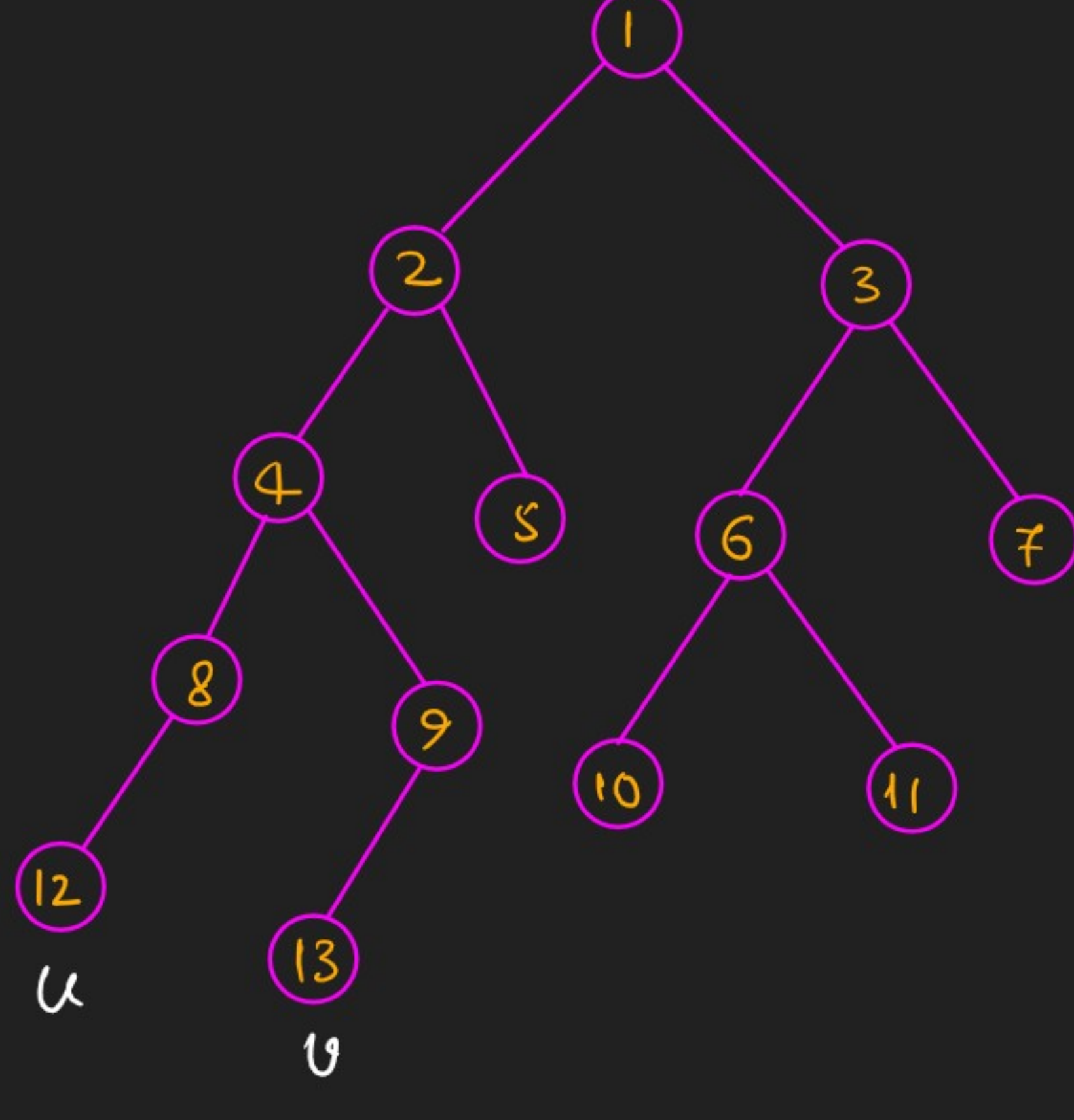


Lowest common Ancestor

Binary lifting



① Ancestor Table

row = n

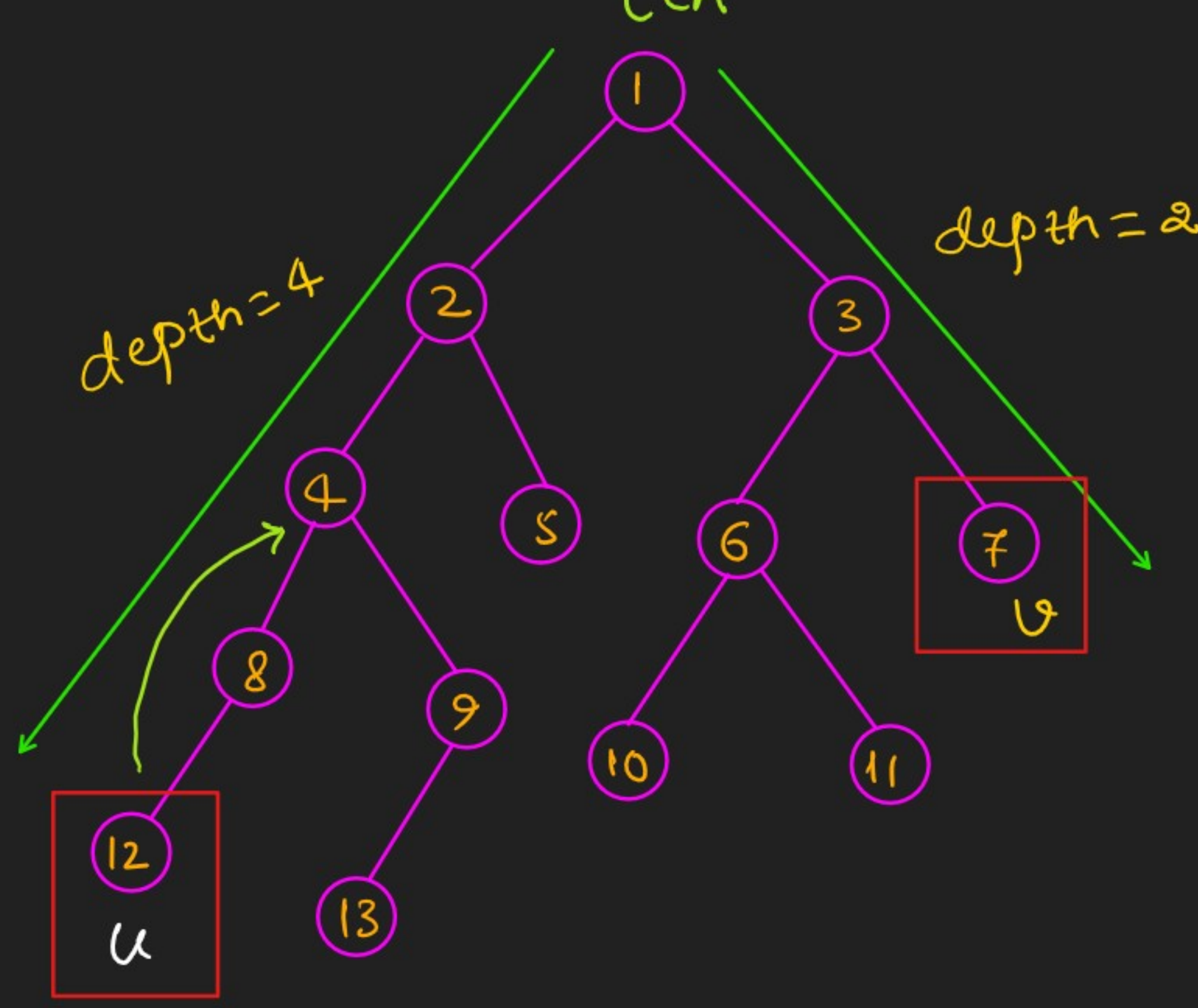
col = $\log_2(n) + 1$

But here $n \rightarrow$ यहाँ पता नहीं है।

```
for(j = cols - 1; j >= 0; j--) {
    if(ancestorTable[u][j] == -1) continue;
    if(ancestorTable[u][j] != ancestorTable[v][j]) {
        u = ancestor[u][j];
        v = ancestor[v][j];
    }
}

// Just below LAC  $\rightarrow u, v$ 
return ancestorTable[u][0];
}
```

* (u, v) in different depths



$\left. \begin{array}{l} \text{depth}(u) = 4 \\ \text{depth}(v) = 2 \end{array} \right\} \text{diff} = 4 - 2 = 2$

k^{th} ancestor of a node

diff = $k = 2 \rightarrow$ यहाँ पता है।

```
for(j = 0; j < cols; j++) {
    if(k and (1 < j)) {
        u = ancestorTable[u][j];
    }
}

for(j = cols - 1; j >= 0; j--) {
    if(ancestorTable[u][j] == -1) continue;
    if(ancestorTable[u][j] != ancestorTable[v][j]) {
        u = ancestor[u][j];
        v = ancestor[v][j];
    }
}

// Just below LAC  $\rightarrow u, v$ 
return ancestorTable[u][0];
}
```

LCA(u, v)

